MODULAR CONTRACTS

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FACTS

Current theory: Contract as incentive/sharing rule: outcome $\mapsto$ transfers

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Real contracts

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Evaluated clause by clause

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<u>Loopholes</u>

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<u>Errors due to complexity</u>

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<u>Incompleteness</u>: Clauses omit relevant contingencies

<u>Conservatism</u>: High burden of truth to verify clause

QUESTIONS

Why are contracts modular, with clauses evaluated separately?

What explains loopholes, complexity, landmines, incompleteness,...?

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Model of modular contracts

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E.g. true data: $\mathbf{a}_j = 110, \mathbf{a}_k = 101$

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Tradeoff: Wide clauses: Susceptible to shuffling; Narrow: Neglect joint info

RESULTS

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Narrow clauses

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Extension: High thresholds

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Extension: High thresholds $\Rightarrow$ conservatism

CONTRIBUTION

Relative to literature on contracting in language (Battigalli–Maggi 02)

Develop framework in which clauses/evaluation matter

Relative to literature on contracting anomalies (e.g. Choi–Gulati–Scott 24)

Develop formal model to rationalize them

MODEL

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(Atomic) Premises

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Target policy: $\mathcal{T} \subset \mathcal{A}$ evaluated on true data

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Evaluation

MODEL: EXAMPLE

Premises: $\mathbf{a}_{\text{IC}} =$ "interest coverage $\geq \kappa_{\text{IC}}$"; $\mathbf{a}_{\text{LR}} =$ "leverage ratio $\leq \kappa_{\text{LR}}$"

Clauses: E.g. $\mathcal{M}_{\text{IC}} = \{\mathbf{a}_{\text{IC}}\}$, $\mathcal{M}_{\text{LR}} = \{\mathbf{a}_{\text{LR}}\}, \mathcal{M}_{\text{joint}} = \{\mathbf{a}_{\text{IC}}, \mathbf{a}_{\text{LR}}\}$

Contract: E.g. $\Phi^{\mathcal{A}} = \{\mathcal{M}_{\text{IC}}, \mathcal{M}_{\text{LR}}\}$ or $\Phi^{\mathcal{T}} = \{\mathcal{M}_{\text{joint}}\}$

Data: Matrix of what premises hold when: E.g. $\left(\mathbf{a}_{\text{IC}}, \mathbf{a}_{\text{LR}}\right) = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$

Shuffled data: Signal preserving marginals: E.g. $\left(\hat{\mathbf{a}}_{\text{IC}}, \hat{\mathbf{a}}_{\text{LR}}\right) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}$

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Target policy

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DEFINITIONS: PREMISE- AND TARGET-BASED

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Premise-based contract: Narrow clauses: $\Phi^{\mathcal{A}} := \left\{\{\mathbf{a}_j\}\right\}_{\mathbf{a}_j \in \mathcal{T}}$

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Target-based contract: Wide clauses: $\Phi^{\mathcal{T}} := \left\{\mathcal{T}\right\}$

BENCHMARKS

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In general: Premise- and target-based contracts coincide: $[\![\Phi^{\mathcal{A}}]\!] = [\![\Phi^{\mathcal{T}}]\!]$

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Evaluation reduces to logic; grouping of premises irrelevant

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In general: Target-based contract implements target policy: $\llbracket \Phi^{\mathcal{T}} \rrbracket = \llbracket \mathcal{T} \rrbracket$

LOOPHOLES

DEFINITION: LOOPHOLE STATES

Loophole states LH: Data realizations s.t. $[\![\Phi^{\mathcal{A}}]\!] \neq [\![\Phi^{\mathcal{T}}]\!]$ even if $e = 0$

EXAMPLE: LOOPHOLE

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Two premises with three data each and $\tau = 2$:

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Two premises with three data each and $\tau = 2$:

i
1
2
3

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Loophole: Evaluation & conjunction don't commute: $[\![\mathbf{a}_j]\!] \wedge [\![\mathbf{a}_k]\!] \neq [\![\mathbf{a}_j \wedge \mathbf{a}_k]\!]$

SHUFFLE RISK

DEFINITION: SHUFFLE RISK STATES

Shuffle risk states SR: Data realizations s.t. $0 < \mathsf{Pr}\left[[\![\Phi^{\mathcal{T}}]\!] = 1 \,\middle|\, \{\mathbf{a}_j\}_j \right] < 1$

EXAMPLE: SHUFFLE RISK

Observe permuted data

i	$\mathbf{a}_j$	$\mathbf{a}_k$	$\mathbf{a}_j \wedge \mathbf{a}_k$
1	1	1	1
2	1	0	0
3	0	1	0
# of 1s $\geq \tau$	**1**	**1**	**0**

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Observe permuted data E.g. $\hat{\mathbf{a}}_j = \mathbf{a}_j$ (no shuffle)

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EXAMPLE: SHUFFLE RISK

Observe permuted data E.g. $\hat{\mathbf{a}}_j = \mathbf{a}_j$ (no shuffle) ; $\hat{\mathbf{a}}_k = 110$ (shuffle of $\mathbf{a}_k$)

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Marginals preserved $\Rightarrow$ premise-based evaluation unchanged

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# of 1s $\geq \tau$	**1**	**1**	**0**	**1**	**1**	**1**

Marginals preserved $\Rightarrow$ premise-based evaluation unchanged

Joint changed: $101 \rightarrow 110 \Rightarrow$ target-based evaluation flips

RESULTS

LEMMA

LEMMA: LOOPHOLES ARE FALSE POSITIVES

If $\llbracket \Phi^{\mathcal{A}} \rrbracket \neq \llbracket \mathcal{T} \rrbracket$, then $\llbracket \Phi^{\mathcal{A}} \rrbracket = 1$

LEMMA: LOOPHOLES FALSE POSITIVES: PROOF

Suppose (for contradiction) that $[\![\mathcal{T}]\!] = 1$ and $[\![\Phi^{\mathcal{A}}]\!] = 0$

$[\![\mathcal{T}]\!] = 1 \Rightarrow \bigwedge_{a_j} a^i_j = 1$ for at least τ "rows"

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$\Rightarrow\Leftarrow$

LEMMA

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK

$\mathsf{LH} \subsetneq \mathsf{SR}$

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK: PROOF

LH $\Rightarrow$ SR

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$\mathsf{LH} \Rightarrow \mathsf{SR}$: Loophole states: Premise-based accepts, target-based rejects

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Premise based accepts if $\min_j \sum_i a_j^i \geq \tau$

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But shuffle could be $\hat{\mathbf{a}}_j = 110, \hat{\mathbf{a}}_k = 101$

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But shuffle could be $\hat{\mathbf{a}}_j = 110, \hat{\mathbf{a}}_k = 101$: $[\![\hat{\mathbf{a}}_j \wedge \hat{\mathbf{a}}_k]\!] = [\![100]\!] = 0$ (yes SR)

R1: LOOPHOLES

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With $e > 0$, if $\mathsf{Pr}[\mathsf{SR} \setminus \mathsf{LH}]$ large, then $\Phi^{\mathcal{A}} \succ \Phi^{\mathcal{T}}$

R1: LOOPHOLES: INTUITION

If loopholes are infrequent, make clauses narrow

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If loopholes are infrequent, make clauses narrow

Accept loopholes to reduce shuffling error

LOOPHOLES IN PRACTICE: J. CREW

J. Crew was distressed and wanted to free up collateral

Credit agreement ostensibly banned operations exploiting creditors, but...

Clause $\mathbf{a}_j$: "Can move assets to restricted sub"

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Show why loopholes persist $e > 0$

R2: ERRORS DUE TO COMPLEXITY

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If $c_{\mathrm{I}}\mathsf{Pr}[\mathsf{LH}]$ sufficiently high, $\Phi^{\mathcal{T}} \succ \Phi^{\mathcal{A}}$

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Empirically: Complexity leads to errors in taxes and regulatory compliance
OECD, Benzarti 20, Zwick 21, Behn et al 22...

R3: REDUNDANCY

DEFINITION: REDUNDANCY

$\Phi^{\mathcal{R}}$ called redundant if there's $\mathbf{a}_j$ s.t. $\mathbf{a}_j \in \mathcal{M}_m \cap \mathcal{M}_{m'}$ for clauses $m \neq m'$

R3: REDUNDANCY

If $e > 0$ not too high and $c_{\mathrm{I}}\mathsf{Pr}[\mathsf{LH}]$ high enough, both $\Phi^{\mathcal{R}} \succ \Phi^{\mathcal{T}}$ & $\Phi^{\mathcal{R}} \succ \Phi^{\mathcal{A}}$

R3: REDUNDANCY: INTUITION

Benefit w.r.t. premise based: Keeps joint information

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Benefit w.r.t. target-based: Protection against shuffling errors

Cost: False negatives

R4: INCOMPLETENESS

DEFINITION: INCOMPLETENESS

$$\Phi^{\mathcal{I}} \text{ called incomplete w.r.t. } \mathcal{T} \text{ if } \mathbf{a}_j \notin \bigcup_{\mathcal{M}_m \in \Phi^{\mathcal{I}}} \mathcal{M}_m \text{ for some } \mathbf{a}_j \in \mathcal{T}$$

R4: INCOMPLETENESS

If e and c_{II} high and $\mathsf{Pr}\left[\llbracket\Phi^{\mathcal{I}}\rrbracket > \llbracket\mathcal{T}\rrbracket\right]$ low, then $\Phi^{\mathcal{I}} \succ \Phi^{\mathcal{T}}$

R4: INCOMPLETENESS

If e and c_{II} high and $\mathsf{Pr}\left[\llbracket\Phi^{\mathcal{I}}\rrbracket > \llbracket\mathcal{T}\rrbracket\right]$ low, then $\Phi^{\mathcal{I}} \succ \Phi^{\mathcal{T}}$

"Sometimes" also $\Phi^{\mathcal{I}} \succ \Phi^{\mathcal{A}}$

R4: INCOMPLETENESS: INTUITION

Long clauses induce shuffling risk $\Rightarrow$ omit some premises to avoid it

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Could use "proxy task" $\mathbf{a}_\ell \approx \mathbf{a}_j \wedge \mathbf{a}_k$ (yielding $\Phi^{\mathcal{I}} \succ \Phi^{\mathcal{A}}$)

R5: CONSERVATISM

EXTENSION

Clause-dependent thresholds $\tau_{\mathcal{M}}$

DEFINITION: CONSERVATISM

Let $\tau_{\mathcal{T}}$ be threshold of target

$\Phi^{\mathcal{C}}$ is <u>conservative</u> if $\tau_{\mathcal{M}_m} > \tau_{\mathcal{T}}$ for every m

R5: CONSERVATISM (INFORMALLY)

"Often" $\Phi^{\mathcal{C}} \succ \Phi^{\mathcal{A}}$ and $\Phi^{\mathcal{C}} \succ \Phi^{\mathcal{T}}$

R5: CONSERVATISM: INTUITION

Premise-based evaluations gives too many false positives

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R5: CONSERVATISM: INTUITION

Premise-based evaluations gives too many false positives

Conservatism decreases them via increased threshold

- Feature: Relies on only marginals $\Rightarrow$ not subject to shuffle risk
- Bug (lite): Still neglects joint info (but fewer false positives than $\Phi^{\mathcal{A}}$)

R5: CONSERVATISM: APPLICATIONS

Law: "beyond reasonable doubt," "reasonable suspicion," "probable cause"
In re Winship, 397 U.S. 358 (1970), Terry v. Ohio, 392 U.S. 1 (1968)

Accounting & auditing: High standard for good news Basu 97; Watts 03

E.g. losses recognized faster than gains

CONCLUSION

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Model of modular contracts in formal language s.t. shuffled data

- Loopholes
- Errors of complexity
- Landmines
- Incompleteness
- Conservatism

MODULAR CONTRACTS